

Skills Summary

Decimal Multiplication & Division

Use this reference while completing your worksheet or when studying.

Skill 1 — Powers of Ten Move the Point

Rule: multiplying by 10, 100, 1000 moves the decimal point 1, 2, 3 places to the *right*; dividing moves it that many places to the *left*. Each digit becomes worth ten times more (or less) per step.

Example: Multiply 2.7×100 .

Step 1. Multiplying by a power of ten only changes what each digit is *worth*, so slide the decimal point — two zeros means two places right.

Step 2. $2.7 \rightarrow 27 \rightarrow 270$ (fill the empty place with a zero)
 $\Rightarrow 2.7 \times 100 = \mathbf{270}$

Try it: Divide $36.5 \div 100$.

check: 0.365

Skill 2 — Multiply, Then Count Decimal Places

Rule: multiply as if both numbers were whole numbers, then give the product as many decimal places as the *two factors have together*. Why: tenths $\times$ tenths = hundredths.

Example: Multiply 0.3×0.7 .

Step 1. Multiply the digits as whole numbers first, then place the decimal point by counting both factors' decimal places.

Step 2. $3 \times 7 = 21$; one place + one place = two places, so the answer is 21 hundredths.
 $\Rightarrow 0.3 \times 0.7 = \mathbf{0.21}$

Try it: Multiply 0.5×0.9 .

check: 0.45

Skill 3 — Divide with a Whole-Number Divisor

Rule: to divide by a decimal, multiply *both* numbers by the same power of ten until the divisor is a whole number. Both numbers shift together, so the quotient does not change.

Example: Divide $3.6 \div 0.4$.

Step 1. Make the divisor whole by shifting both numbers one place — the quotient stays the same because both grew tenfold.

Step 2. $3.6 \div 0.4 = 36 \div 4$
 $\Rightarrow 3.6 \div 0.4 = \mathbf{9}$

Try it: Divide $5.6 \div 0.8$.

check: 7

(!) Watch out: an answer can look ten times too big or too small and still have the right digits. *Estimate first* (round each number), then check that your exact answer is close to the estimate.